Reconstructing gaps in coastal sensor records

Benchmarking imputation of chlorophyll-*a* and dissolved oxygen in Tongoy Bay, Chile

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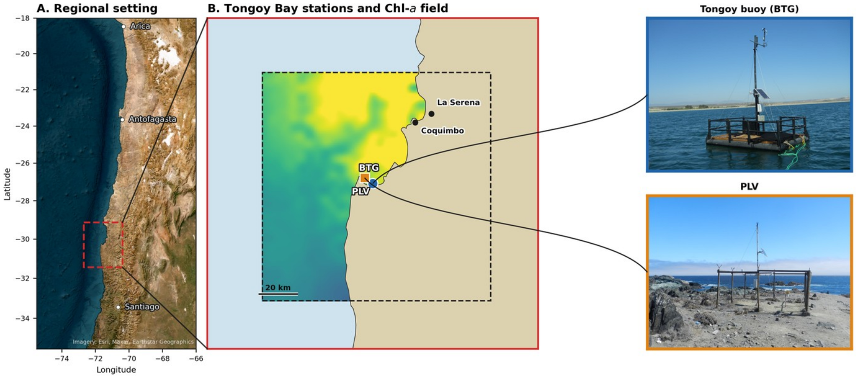
Objective: a reusable workflow for reconstructing gaps in coastal sensor records.

Long coastal records are valuable but incomplete



Two sensors on the Tongoy buoy record chlorophyll-a and dissolved oxygen over 2015–2026.

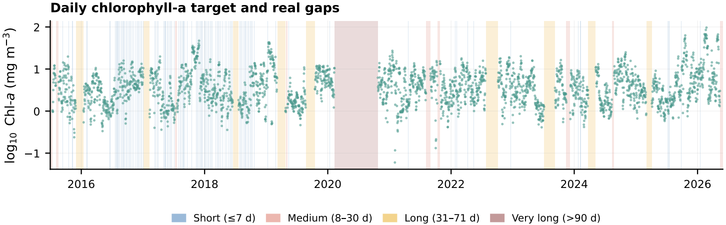
The record is interrupted by real gaps

24.5%

chlorophyll days missing

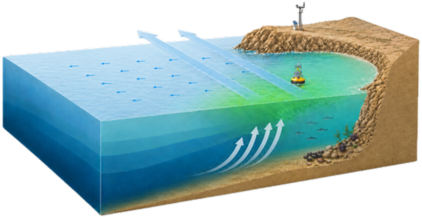
27.8%

oxygen days missing



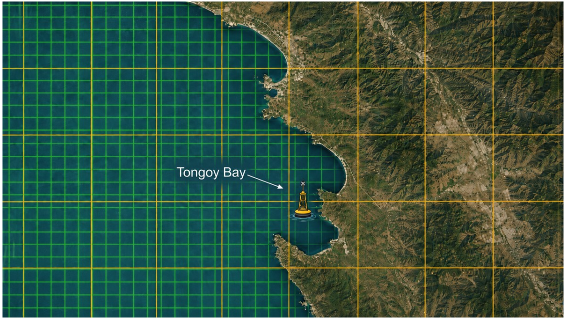
Reconstruct interrupted historical records, keeping real gaps separate from validation evidence.

Missing values sit in a dynamic, multiscale system



Wind-driven offshore transport and coastal upwelling near Tongoy.

Montecino & Lange (2009).



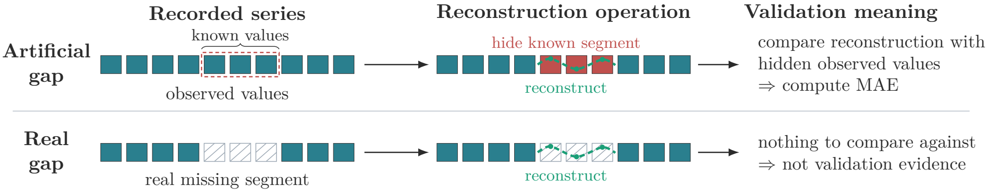
The buoy is a nearshore point; external satellite products are coarser grid fields.

Skill is measured only where truth is withheld

Artificial gap. Hide known observations, reconstruct them, and score against the withheld truth.

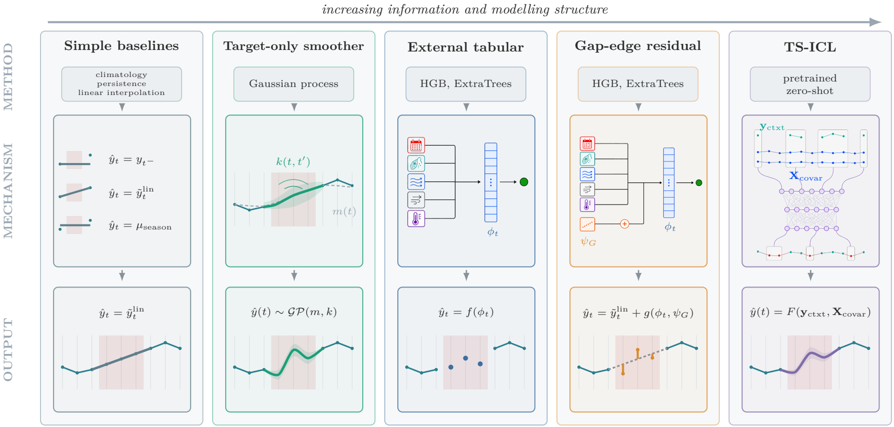
Real gap. Reconstruct values that were never observed — no error can be computed.

> Real-gap reconstructions are **outputs**, not validation evidence.



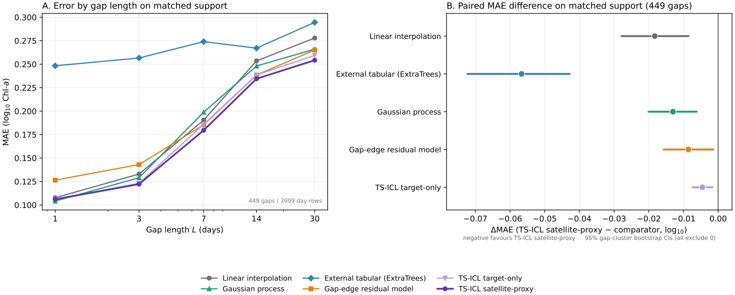
Methods are ranked *only* on artificial gaps — for chlorophyll and for oxygen alike.

One task, five information structures



All five groups represent temporal context differently and are ranked on the same benchmark.

Chlorophyll: TS-ICL edges past a strong baseline

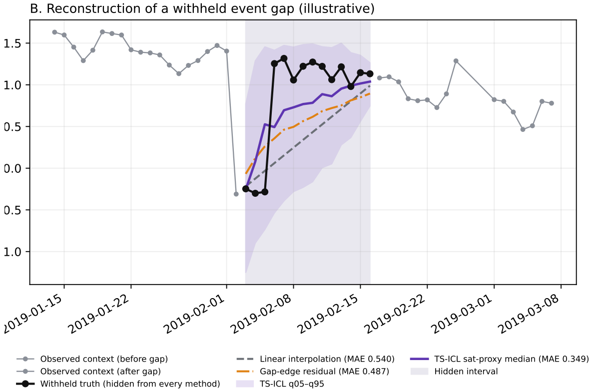


0.239	0.221	0.277
interpolation MAE ($\log_{10}$)	TS-ICL + satellite Chl a *(best)*	ExtraTrees, external-only

Wind carries information; blooms stay hard

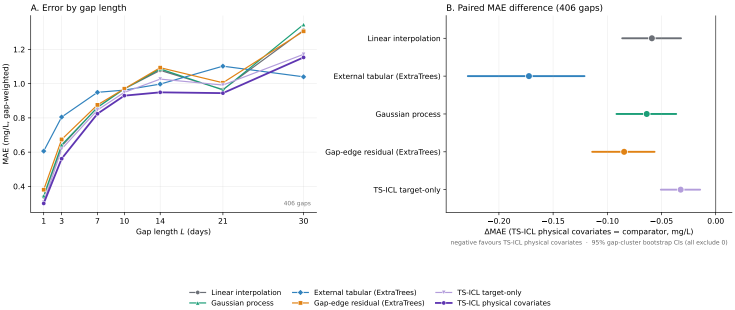
Covariate signal (a family counts only if it beats all four disrupted placebos):

Covariate family	Beats placebos
Wind / upwelling	✓ 4/4
Curated physical set	✓ 4/4
Solar radiation	∼ 2/4
SST / thermal	× 1/4
Ocean currents	× 0/4



Every method has higher error on event gaps ($\geq$1 day above Chl a p90).

Oxygen: the same workflow transfers



0.733

interpolation MAE (mg/L)

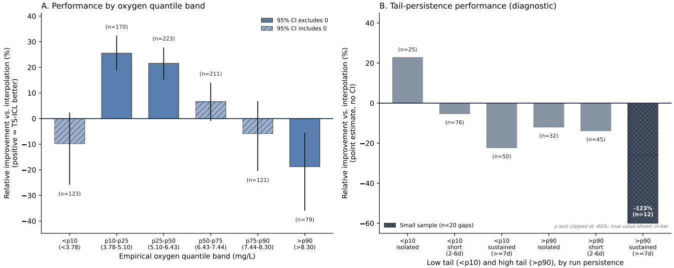
0.675

TS-ICL + physical bundle *(best)*

0.847

external tabular: 15.5% higher MAE

The gain disappears in the tails



- Gains concentrate in the **p10–p50 middle**.
- TS-ICL is **worse than interpolation in both tails**.
- **Sustained** episodes are hardest (small strata — read cautiously).

Pooled MAE can **hide failure** where it matters most.

The workflow transfers; the failure regimes don't

Chlorophyll-*a*

Best arm	TS-ICL satellite-proxy
Useful signal	wind / upwelling
Hard regime	high-Chl *a* events

Dissolved oxygen

Best arm	TS-ICL physical cov.
Useful signal	currents
Hard regime	distribution tails

Shared conclusions. **1.** Interpolation is a strong reference. **2.** External predictors alone are insufficient. **3.** Test covariates empirically. **4.** Regime diagnostics are required alongside pooled metrics.

The transferable object is the **benchmark workflow itself** — the physical signal and failure regime are target-specific.

Limitations

Validation and model limits

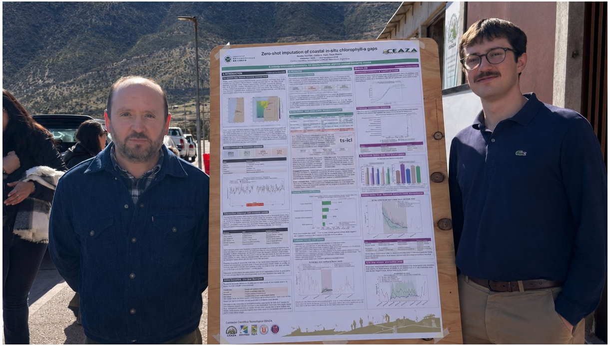
- Real outages such as the 256-day gap exceed the validated gap lengths.
- Only one pretrained sequence architecture was tested.
- Event-day and tail-regime uncertainty calibration remain incomplete.

Physical and scope limits

- External products are coarser than the buoy-scale signal.
- Covariate evidence is correlational, not causal.
- Validation is single-station; generality across CEAZA stations is untested.

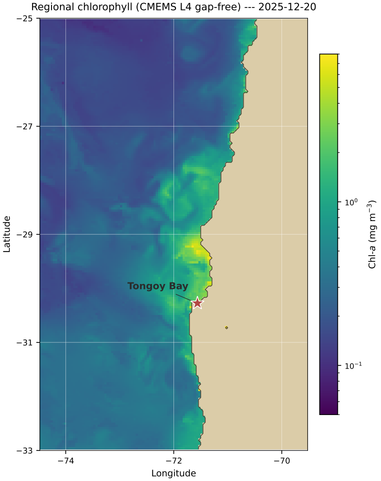
Validate against strong baselines, preserve physical context, and examine where average skill fails.

Presented to the CEAZA research community



Presented as a CEAZA poster in La Serena.

Current work: regional ocean-colour forecasting



- From point-sensor reconstruction to **regional spatial prediction**.
- Forecasts near-future chlorophyll fields from recent satellite ocean-colour and physical forcing over the shelf, 25–33°S.
- Conceived as a **practical forecasting service**.

Regional CMEMS L4 gap-free chlorophyll, 2025-12-20, shown for context.

Backup — Daily target definition, QC and missingness

A day is eligible only if ≥ 18 valid hourly values ($\geq 75\%$ coverage). Span 2015-07-01 to 2026-05-31 (3988 possible daily records).

Target	Channel / unit	Eligible	Missing	Missing %	Real gaps
Chlorophyll-a	BTGCLF, $\mathrm{mg\,m^{-3}}$ ($\log_{10}$)	3012	976	24.5%	128
Dissolved oxygen	BTGOXD2, mg/L	2880	1108	27.8%	125

Distribution (eligible days)	Median	IQR	p05	p95
Chlorophyll-a ($\mathrm{mg\,m^{-3}}$)	3.48	5.04	0.79	15.93
Dissolved oxygen (mg/L)	6.43	2.34	2.82	8.80

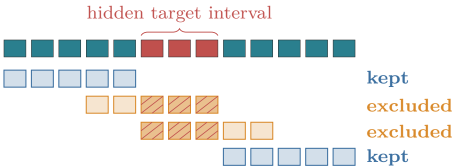
Chlorophyll is right-skewed $\Rightarrow$ modelled on $\log_{10}$; oxygen modelled directly in mg/L.

Backup — Feature families and leakage control

Feature family	Target
Calendar / seasonality	both
Target history	both
Gap-edge context	both
Satellite biological proxy	Chl *a*
Wind / upwelling forcing	both
Solar / atmospheric forcing	both
SST / thermal state	both
Current / transport	both
Local BTG diagnostics	oxygen

Tabular arms see one feature vector per day; TS-ICL sees the same information as time-indexed channels.

Dependency-window leakage control



A tabular row is dropped when its target-derived window overlaps the hidden interval. External covariates stay available.

Backup — Chlorophyll benchmark tables

Pooled MAE ($\log_{10}$), 449 gaps

Method	MAE
TS-ICL satellite-proxy	**0.221**
TS-ICL target-only	0.225
Gap-edge residual	0.229
Gaussian process	0.234
Linear interpolation	0.239
External tab. (ExtraTrees)	0.277
External tab. (HGB)	0.287

MAE by gap length L

L	Interp	GP	Ext	Edge	TS tgt	TS sat
1	0.108	0.104	0.248	0.127	0.106	0.106
3	0.133	0.129	0.257	0.143	0.123	0.122
7	0.190	0.199	0.274	0.185	0.186	0.180
14	0.254	0.248	0.267	0.239	0.239	0.235
30	0.278	0.266	0.295	0.265	0.259	0.254

Event / non-event MAE (TS-ICL sat): 0.199 / 0.266; external tabular worst on events (0.235 / 0.366).

TS-ICL satellite-proxy quantile calibration (681-gap pool): 50%→51.3%, 80%→81.8%, 90%→91.6%.

Backup — Oxygen MAE by gap length

L	n	Interp	Ext	GP	Edge	TS tgt	TS phys
1	100	0.339	0.606	0.345	0.380	0.307	0.300
3	100	0.635	0.805	0.642	0.674	0.618	0.562
7	78	0.863	0.949	0.861	0.876	0.846	0.824
10	55	0.970	0.962	0.970	0.971	0.949	0.930
14	36	1.078	0.998	1.086	1.094	1.027	0.949
21	22	0.966	1.102	0.963	1.005	0.992	0.945
30	15	1.312	1.040	1.346	1.306	1.171	1.153

All values are gap-weighted MAE in mg/L. n = number of gaps at that length. Pooled: interpolation 0.733, TS-ICL physical 0.675 (−8.0%).

Backup — Oxygen paired comparisons

Comparator	ΔMAE vs TS-ICL physical	95% CI
Interpolation	−0.059	[−0.086, −0.032]
Ext. tabular	−0.172	[−0.229, −0.121]
Gaussian process	−0.064	[−0.091, −0.036]
Gap-edge	−0.084	[−0.114, −0.056]
TS-ICL target	−0.032	[−0.051, −0.014]

Negative ΔMAE favours TS-ICL physical covariates. All five comparators' 95% gap-cluster bootstrap CIs exclude zero. Pooled: interpolation 0.733, TS-ICL physical 0.675 (−8.0%).

Backup — Oxygen tails and the L=30 audit

Relative improvement by oxygen quantile band

Band (mg/L)	n	Rel. impr. [95% CI]
<p10 (<3.78)	123	−9.8% [−25.7, +2.2]
p10–p25 (3.78–5.10)	170	+25.5% [+19.0, +32.2]
p25–p50 (5.10–6.43)	223	+21.6% [+15.3, +27.6]
p50–p75 (6.43–7.44)	211	+6.7% [−0.8, +13.9]
p75–p90 (7.44–8.30)	121	−5.9% [−20.3, +6.7]
>p90 (>8.30)	79	−18.8% [−35.8, −5.6]

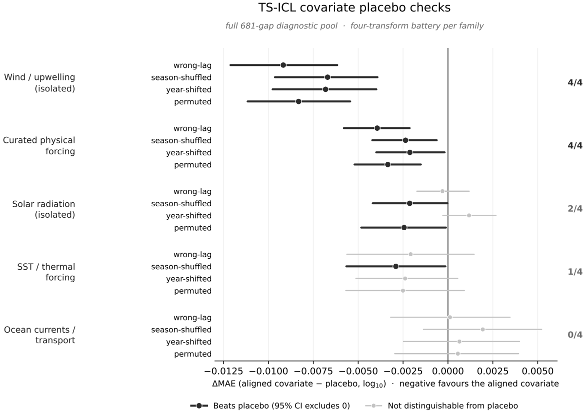
Sustained high tail (≥7 d, n=12): −123% — small stratum.

L=30 **external-tabular reversal — diagnostic only**

- Only 15 gaps in the stratum.
- Paired-difference 95% CI [−0.644, +0.092] **includes 0**.
- Driven by a few high-error interpolation cases.

Kept as a small-stratum caveat — **not** a headline reversal.

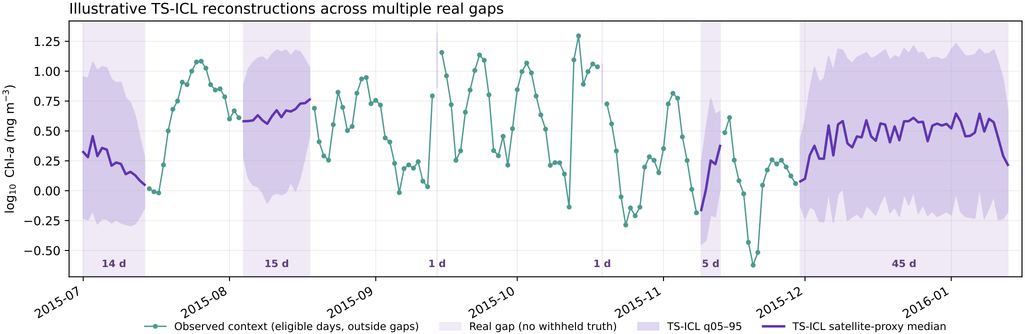
Backup — TS-ICL covariate placebo battery (chlorophyll)



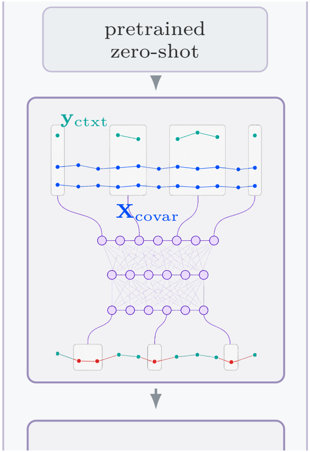
A family counts only if it beats all four placebos, CI excluding zero.

Backup — Real-gap reconstructions (illustrative only)

Real gaps: no withheld truth; not validation evidence



Backup — TS-ICL input and architecture



- Visible target context $\mathbf{y}_{\text{ctxt}}$ plus the masked hidden interval form one timestamp–value sequence; covariates $\mathbf{X}_{\text{covar}}$ enter as time-indexed channels.
- Timestamp–value pairs and covariates are encoded, then **aggregated across channels** into timestamp-specific context representations.
- An **in-context regressor** (causal self-attention) reconstructs the hidden interval: $\hat{\mathbf{y}}_G = F(\mathbf{y}_{\text{ctxt}}, \mathbf{X}_{\text{covar}})$.
- Output includes **quantile predictions** (q05–q95), not just a point estimate.
- Used **zero-shot**: pretrained elsewhere, no Tongoy-specific fitting.

TS-ICL: Le Naour, Nabil & Petralia (2026); software: EDF Lab (2026).

Backup — Artificial-gap design and validation support

Target	Support	Gap lengths	Gaps	Role
Chlorophyll	full TS-ICL pool	1, 3, 7, 10, 14, 21, 30, 45, 60	681	calibration/placebo
Chlorophyll	shared ladder support	1, 3, 7, 14, 30	449	main ranking
Oxygen	primary support	1, 3, 7, 10, 14, 21, 30	406	main + tails
Oxygen	exploratory longer	>30	6	exploratory only

- Gaps are stratified by **season and year** where the observed record allows it.
- Uncertainty uses a **gap-cluster bootstrap**: the resampling unit is the whole gap, not the individual day, so within-gap errors are not treated as independent.
- Longer gaps are rarer because they require long uninterrupted **observed** runs to construct — this is why claims beyond L=30–60 are not part of the headline ranking.

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